Amirmohammad (Amir) Mohammadi
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Google Scholar | LinkedIn | Website | U.S. Permanent Resident

RESEARCH INTERESTS

Machine Learning, Deep Learning, Foundation Models, Parameter-Efficient Fine-Tuning, Computer Vision, Physics-Informed Machine Learning, Multi-Modal Learning, Generative AI, Biomedical Sensing and Imaging.

TECHNICAL SKILLS

Coding & Frameworks: Python, NumPy, Pandas, Matplotlib, PyTorch, Scikit-Learn, Hugging Face.
Development & Tools: VS Code, Jupyter Notebook, Git, Linux, High-Performance Computing.

EDUCATION

Texas A&M University, College Station, Texas — September 2022 - December 2026
Doctor of Philosophy in Computer Engineering
Thesis: Distribution-guided Parameter-efficient Fine-tuning of Foundation Models for Waveform-based Sensing

Sharif University of Technology, Tehran, Iran — September 2018 - February 2021
Master of Science in Electrical Engineering
Thesis: Design and Evaluation of an Integrated Biomedical System for Human Stress Detection and Monitoring

University of Tabriz, Tabriz, Iran — September 2014 - September 2018
Bachelor of Science in Electrical Engineering
Project: Development of a Data Logger for Motion Sensors with Python

EXPERIENCE

Texas A&M University, College Station, Texas — September 2022 – December 2026
Graduate Research Assistant

- **Developed** a distribution-aware method using histogram layers for parameter-efficient fine-tuning of Transformer-based foundation models for waveform-based sensing applications.
- **Built** and evaluated a few-shot whole-slide image classification pipeline to investigate parameter-efficient fine-tuning of vision-language models to predict ethanol drinking status from liver histology.
- **Contributed** to an evaluation pipeline for image classification, semantic segmentation, and object detection, comparing pre-trained computer vision models and adaptation strategies.
- **Improved** the classification accuracy of a time-delay convolutional neural network by designing a time-frequency feature combination for acoustic applications.
- **Reduced** ground truth data requirements by a factor of 15 for physiological signals by developing physics-informed neural networks for cuffless blood pressure measurement.
- **Conducted** research supported by fully funded assistantship from MIT Lincoln Laboratory, the National Institutes of Health, and the Texas A&M University Department of Electrical & Computer Engineering.

PEER-REVIEWED PUBLICATIONS

- **A. Mohammadi,** R. Khadka, K. A. Grant, M. L. Benton, and J. Peeples, "Predicting Ethanol Drinking Status from Nonhuman Primate Liver Histology: A Proof of Concept for Parameter-efficient Fine-tuning

in Few-Shot Vision-Language Modeling," *IEEE International Conference on Bioinformatics and Biomedicine,* 2026. (Accepted)

- **A. Mohammadi**, D. Carreiro, A. Van Dine, and J. Peeples, "Histogram-based Parameter-efficient Tuning for Passive and Active Sonar Classification," *IGARSS 2026 - 2026 IEEE International Geoscience and Remote Sensing Symposium*, Washington DC, DC, USA, 2026, pp. 5031-5035. [link].
- J. Ritu, **A. Mohammadi,** D. Carreiro, A. V. Dine, and J. Peeples, "Structural and Statistical Audio Texture Knowledge Distillation for Acoustic Classification," in *IEEE Access*, vol. 14, pp. 145754-145771, 2026. [link]
- F. Orvati Nia, **A. Mohammadi**, S. Al Kharsa, P. Naikare, Z. Hampel-Arias, and J. Peeples, "Neighborhood Feature Pooling for Remote Sensing Image Classification," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*, 2026, pp. 1393–1402. [link]
- **A. Mohammadi**, I. Masabarakiza, E. Barnes, D. Carreiro, A. Van Dine and J. Peeples, "Investigation of Time-Frequency Feature Combinations with Histogram Layer Time Delay Neural Networks," *OCEANS 2025 Brest*, BREST, France, 2025, pp. 1-6. [link]
- **A. Mohammadi**, T. Kelhe, D. Carreiro, A. Van Dine, and J. Peeples, "Cross-Domain Knowledge Transfer for Underwater Acoustic Classification Using Pre-trained Models," in *OCEANS 2025 Brest*, BREST, France, 2025, pp. 1-6. [link]
- K. Sel, **A. Mohammadi**, R. I. Pettigrew, and R. Jafari, "Physics-informed neural networks for modeling physiological time series for cuffless blood pressure estimation," *npj Digital Medicine*, vol. 6, no. 1, p. 110, 2023. [link]
- **A. Mohammadi**, M. Fakharzadeh, and B. Baraeinejad, "An Integrated Human Stress Detection Sensor Using Supervised Algorithms," in *IEEE Sensors Journal*, vol. 22, no. 8, pp. 8216-8223, 2022. [link]

MANUSCRIPTS UNDER REVIEW

- **A. Mohammadi,** A. Van Dine, and J. Peeples, "Parameter-efficient Dual-encoder Architecture with Differentiable Choquet Integral Fusion for Underwater Acoustic Classification". [link]

ACADEMIC SERVICE, TEACHING & MENTORING

- Reviewer, IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP).
- Session chair, IEEE International Geoscience and Remote Sensing Symposium (IGARSS).
- Delivered guest lectures in Machine Learning for Electrical Engineering at Texas A&M University.
- Mentored undergraduate and graduate students on end-to-end AI projects.